CS368 Project

Soil moisture sensor

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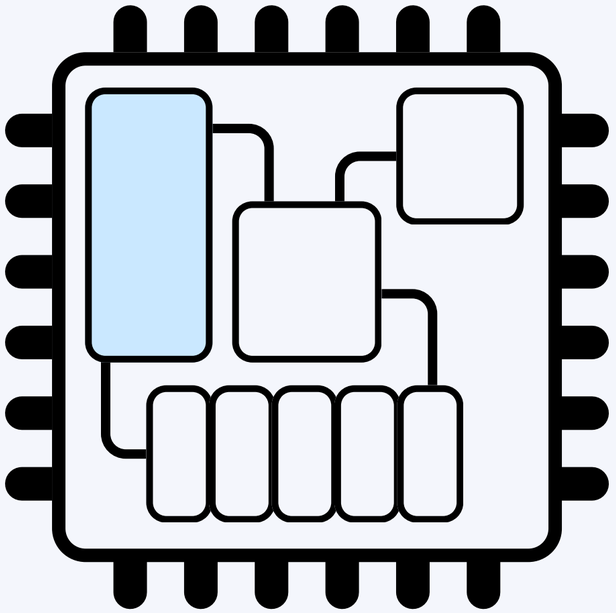
What the paper did

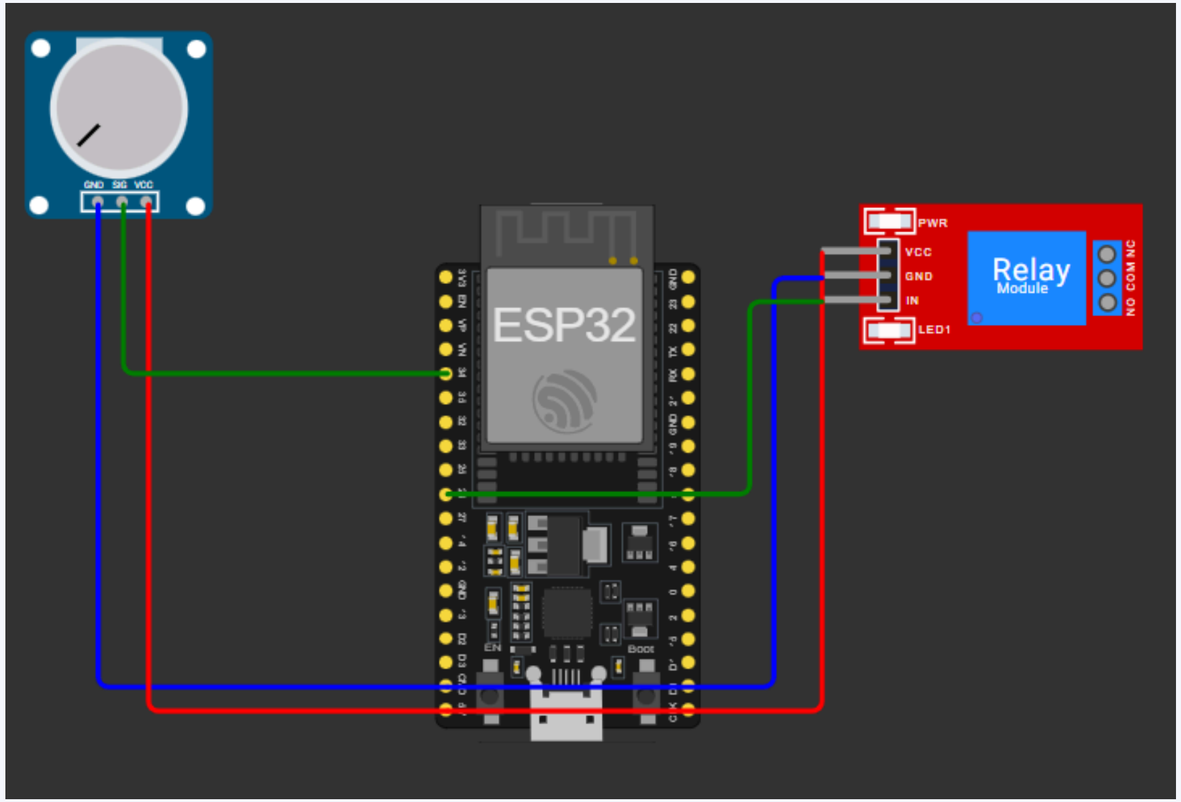
- Proposed a smart irrigation system using IoT and Machine Learning
- Used soil moisture sensor as a key input for irrigation decisions
- Collected sensor data using ESP32 microcontroller
- Trained a supervised ML model (KNN) on historical data
- ML model predicts irrigation ON or OFF
- Aim was water conservation and automation

Our Approach

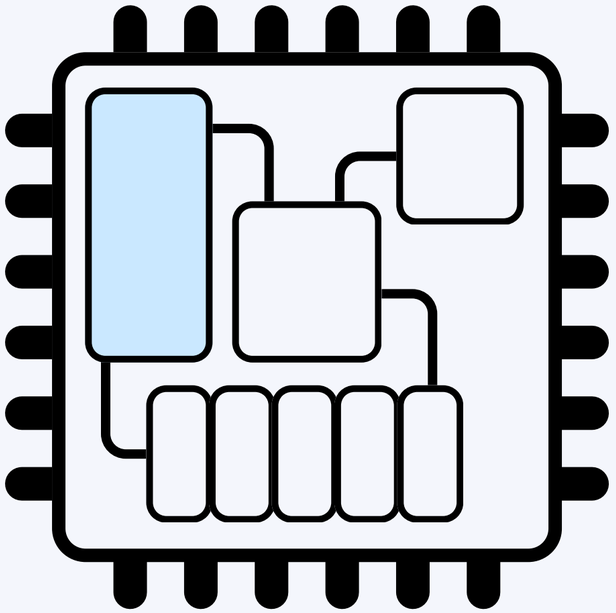
- Focus only on the soil moisture part of the system
- Read soil moisture values using ESP32 (simulation)
- Collect moisture readings over time
- Store readings as a dataset (CSV format)
- Train a machine learning model using soil moisture data
- Use the model to predict irrigation requirement



Components



- ESP32 microcontroller
- Capacitive soil moisture sensor
- (simulated using potentiometer in Wokwi)
- Laptop / PC for:
  - Data storage
  - ML model training
- Software tools:
  - Wokwi (simulation)
  - Python (ML and data analysis)

Future Plan

- Add relay and pump control for real irrigation
- Include temperature and humidity sensors
- Deploy system on real ESP32 hardware
- Improve ML accuracy with more data
- Use cloud storage for remote monitoring
- Extend system for full smart irrigation automation